The 043 Datasets

Fifteen expression, protein and morphology datasets: genotypes, agreement, co-location and candidate training batches

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Abstract

The 043 store holds 173,617 records of 15 yeast datasets that read mRNA, protein abundance or turnover, or CalMorph morphology, kept as served with no aggregation. Its genotypes fall into four classes, and today a model sees only one of them as intended: deletions are gene sets, isolates mark almost every gene as perturbed, segregants mark none, and the promoter library has no gene-level genotype. Each dataset is set against its own reproducibility. Kemmeren 2014's two cultures of 114 mutants give an own-culture ceiling of 0.729; it replaces the experiment 019 ceiling, which rested on Sameith 2015's single mutants, found here to be Kemmeren's own arrays. Two constructions of one Jackson 2020 genotype agree on the deletion effect at a median r of 0.129 against 0.095 for different genotypes; Hu 2007 released no replicate; no segregant appears in two panels. Between datasets, agreement is low (median per-genotype r from -0.006 to 0.208 apart from the Sameith copy), and morphology is the label most often shared with expression and protein. A proposal for two training batches, deletions with morphology held out and isolates with segregants under a stated allele assumption, closes the document; it is presented for decision, not decided.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The 043 store ✗

Query. The store is one query over the served knowledge graph, KG 4.0 (release `2026.10.17-4295f7ca`), registered as the supported query `expression_proteome_morphology_043`. It takes every served yeast dataset whose readout is a vector of mRNA expression, protein abundance or turnover, or CalMorph morphology: 15 datasets, 173,805 served experiments. Five candidates were left out: Puddu 2019 holds a realized-genotype table rather than a cell-state readout, Ohnuki 2018 and 2022 share no strain with the other sets, Hackett 2020 carries an induction the graph processor would read as a deletion, and Messner 2021 carries no genotype perturbation.

The keep-the-data rule. The store holds one sample per served pair of experiment and experiment reference, with no unit converter, no deduplicator and no aggregation over genotypes. Two reasons were measured on the code: the mean deduplicator keys on the perturbed gene names alone, so it would average one deletion across Jackson 2020's eleven conditions, and it raises on four of the store's experiment types. Reconciling units and replicates is therefore a modeling decision, taken per batch, rather than a property of the store.

Size. The raw store holds 173,617 samples. The 188 served experiments it lacks name a gene outside the genome's 6,607-gene set and fail the deletion blocks' clause that every perturbed gene be in that set: Kemmeren 2, Hu 1, Nadal-Ribelles 70, Messner 75, Ohya 40 (slurm job 5855; `results/store_census.json`).

Terms. Table 1 fixes the words the rest of the text uses. Table 2 gives each dataset with the statistic that bounds what a model can reach on it.

Table 1: Controlled vocabulary, alphabetical.

Term	Meaning
Ceiling	The per-gene Pearson a perfect predictor of the true signal would score against one measurement: $\sqrt{\text{rel}_g}$, averaged over genes, with rel_g the share of variance across strains that is signal. It is a bound on a model, not a score of one.
Construction	One independently built strain of a genotype. Jackson 2020 built six per genotype, each with its own transcribed barcode, and the store keeps each as its own record.
Co-location	Two labels measured on the same genotype key, in any datasets.
Deletion-effect profile	Per gene, $\log_2((\text{CPM}+1)/(\text{CPM}_{\text{HO}}+1))$ for a construction against the HO-deletion control of the same condition.
Haplotype block	A run of markers on one chromosome over which a segregant's called parent and that call's posterior are constant; the genotype of a segregant is a list of blocks against two pinned parent assemblies.
Isolate	A natural strain; its genotype is thousands of sequence variants, gene presences and absences, and natural loss-of-function calls against the reference.
Pseudobulk	The UMI counts of several single cells summed per gene.
Segregant	A haploid meiotic product of a cross between two parent strains. Pooled segregants were genotyped by sequencing before profiling; de novo segregants are genotyped from each cell's own transcripts.
Test-retest	Two independent measurements of the same strains, correlated across strains per gene (r_g, so $\text{rel}_g = r_g$) or across genes per strain.

Table 2: The 15 datasets of the 043 store. Genotypes and records count the raw store; features per record is the median length of the stored label vector (genes, proteins or CalMorph features; one scalar for Vaishnav). Environments count distinct media, temperature and environmental perturbation; temperature is a typed gap where the paper states none. The last column gives each dataset's agreement with itself and the statistic used: a ceiling is the mean over genes of $\sqrt{\text{rel}_g}$ (Sec. 3.1); Jackson is the median Pearson between two constructions of one genotype against two of different genotypes, Nadal-Ribelles the median Pearson of one genotype across batches against different genotypes. Messner, Zelezniak, Ohya, Caudal and Nadal-Ribelles were measured in experiments 019 and 028, before this store.

Dataset	Genotype class	Readout, scale	Genotypes, n	Records, n	Features per record, median	Environments, n: name	Within-dataset agreement, statistic in cell
Kemmeren 2014	deletion	two-color array, log2 ratio to WT pool	1,482	1,482	6,169	1: SC, 30 C	0.73 own-culture ceiling, per gene (114 mutants with 2 cultures)
Sameith 2015 Sm	deletion	two-color array, log2 ratio to WT pool	82	82	6,169	1: SC, 30 C	none: the same arrays as Kemmeren (82 of 82 identical)
Sameith 2015 Dm	double deletion	two-color array, log2 ratio to WT pool	72	72	6,169	1: SC, 30 C	not measured
Hu 2007	deletion	error-model X score vs WT, signed	262	262	6,381	1: YPD, temperature a typed gap	not measurable: replicates averaged before release; Kemmeren × Hu r 0.11 is the bound
Jackson 2020	deletion, 6 construc-tions	scRNA-seq UMI counts, summed per construction	12	784	4,028	10: 9 media	construction pair r 0.13 vs 0.09 null, effect, floor 100
Nadal-Ribelles 2025	deletion	pseudobulk log2 fold change vs WT	3,059	6,118	5,866	2: 1 media	same genotype across batches r 0.043 vs 0.022 null
Messner 2023	deletion	SWATH protein abundance, linear	4,476	4,624	1,819	1: SM, 30 C	0.70 ceiling, 388 WT replicates
Zelezniak 2018	deletion	SWATH protein abundance, log	97	97	726	1: SM, 30 C	0.92 ceiling, per-strain replicate SE
Ohya 2005	deletion	CalMorph features, per feature	4,678	4,678	281	1: YPD, 25 C	0.61 ceiling, 122 WT replicates
Caudal 2024	isolate	RNA-seq TPM, linear	943	943	6,050	1: SC, 30 C	0.60 ceiling, 29 re-cultured isolates
Muenzner 2024 protein	isolate	SWATH protein abundance, linear	795	795	1,540	1: SM, 30 C	not measured
Muenzner 2024 turnover	isolate	SILAC degradation rate, per hour	55	55	1,055	1: SM + 80 mg/L [13C6/15N2] L-lysine, 30 C	not measured
Boocock 2025	segregant	scRNA-seq UMI counts, per cell or per segregant	79,516	79,516	962	2: SM, 30 C; 2x YNB + 2% glucose added to FACS sheath fluid, 30 C	no segregant in two panels; nearest-genotype deviation r 0.11 vs 0.07 random
N'Guessan 2025	segregant	scRNA-seq UMI counts, summed per segregant	2,951	2,951	2,592	1: YPD, temperature a typed gap	not measurable: cells summed per segregant, no replicate
Vaishnav 2022	promoter library	sort-seq expression level, one scalar	1	71,158	1	1: SC-Ura, 30 C	not measured
Total				173,617			

2 Genotype classes and how each enters a model ✗

The store holds four genotype classes, and the graph processors that turn a record into model input treat them differently. What follows is the behavior of the code at the 043 build (`torchcell/data/graph_processor.py`), measured on the processed samples.

Deletions. Kemmeren, Sameith, Hu, Jackson, Nadal-Ribelles, Messner, Zelezniak and Ohya: every record names one or two deleted genes (two for Sameith Dm). The genotype is the set of systematic names, and the `Perturbation` processor marks those nodes in `pert_mask`. Jackson 2020 adds a construction identity (plate well and barcode) below the gene name, so its 784 records hold 72 constructions of 12 genotypes.

Isolates. Caudal 2024 and the two Muenzner 2024 datasets: a natural isolate's genotype is its variant calls against S288C, five perturbation types with the counts of Table 3, between 195 and 7,025 perturbations per Caudal record. The `Perturbation` processor marks every gene with any of them, so a Caudal sample has 5,575 of 6,607 nodes in `pert_mask`: the mask then separates almost nothing, and a rule over perturbation types (for example loss of function and absence only) is needed before isolates are trained on.

Table 3: Perturbations of an isolate record, by type. Mean count per record over the records of each dataset (`results/store_census.json`). Sequence variants are named by the gene they fall in; natural loss of function, gene absence and gene presence are released calls; the few deletions are engineered markers.

Dataset	Records, n	Perturbations per record, min to max	Sequence variant, mean per record	Natural LoF, mean per record	Gene absence, mean per record	Gene presence, mean per record	Deletion, mean per record
Caudal 2024	943	195 to 7,025	5,047	383	135	84	0
Muenzner 2024 protein	795	0 to 7,025	5,017	377	133	84	0
Muenzner 2024 turnover	55	4,903 to 6,430	5,059	373	134	90	0

Segregants. Boocock 2025 and N'Guessan 2025: a `SegregantGenotype` names a cross, two parent strains pinned to sha256-verified assemblies, and a list of haplotype blocks, and it carries no perturbation list by design. Since the stage-2 fix the processors read a missing list as zero perturbed genes, so a segregant sample today has `pert_mask` all false: a model sees it as the reference strain. Using segregants needs a representation of the blocks; Sec. 5 names one.

Promoter library. Vaishnav 2022: each of 71,158 records is one 80 bp variable promoter insert driving a YFP reporter, read as one scalar. Every record names the same reporter, so the store holds one gene-name genotype for all of them; the perturbation sets separate them only by sequence content. No gene-level vector and no gene-set encoder applies.

3 Agreement within and between datasets ✗

3.1 Within a dataset ✗

A dataset's agreement with itself bounds both what a model can score on it and what any other dataset can agree with it. The new measurements are in `experiments/043-expression-proteome-morphology/scripts/within_dataset_ceilings.py` (slurm jobs 5893 for Kemmeren and Hu, 5895, 5896 and 5898 for Jackson, the segregants and the figures).

Jackson 2020. Each of the 784 records is one construction in one condition, a median of 23 cells and 43,851 UMIs (IQR 19,757 to 101,152); YPD and rapamycin reach a median of about 400,000 UMIs per construction, the other nine conditions 14,000 to 93,000. Within one condition, two constructions of the same genotype agree on their deletion-effect profiles only slightly better than two constructions of different genotypes (Table 4, Fig. 1a): a median r of 0.129 against 0.095 with every gene of at least 100 UMIs in the condition (1,778 same-genotype and 21,300 different-genotype pairs). The gap widens on the 660 deepest genes (0.244 against 0.103 at 1,000 UMIs) and when only genes covered in both constructions are kept, but that second rule scores fewer pairs, the deep ones. The different-genotype pairs are not near zero because both constructions are divided by the same HO reference. The Jackson ceiling for one construction's deletion effect is therefore low; most of a record's deviation from HO is count noise shared with any other construction in the flask.

Kemmeren 2014. GEO titles name the culture: `bck1-del-1-a` and `-1-b` are the dye-swapped arrays of culture 1. Of 1,480 mutants whose titles parse, 114 carry two cultures, 72 in the responsive series GSE42527 and 42 in the non-responsive series GSE42526; the other 1,366 have one (16 arrays with irregular titles are left out). Culture 1 against culture 2 of the same mutant gives a median per-strain r of 0.730, against 0.284 for culture 1 of one mutant and culture

Table 4: Jackson 2020, construction against construction within one condition. Pooled over the 11 conditions. Level profile: Pearson over genes of $\log_2(\mathrm{CPM}+1)$; deletion effect: of $\log_2((\mathrm{CPM}+1)/(\mathrm{CPM}_{\mathrm{HO}}+1))$ against the six HO constructions summed, HO pairs left out. A gene enters every pair of a condition when its UMIs summed over that condition reach the floor. Same genotype: two of the six constructions of one genotype; different: two constructions of different genotypes in the same flask.

Gene rule	Level profile r, median, same vs different genotype	Deletion-effect r, median, same vs different genotype	Pairs, n, same / different	Genes per pair, median
$\geq$ 10 UMIs summed over the condition	0.637 vs 0.620	0.184 vs 0.157	1,778 / 21,300	5,405
$\geq$ 100 UMIs summed over the condition	0.654 vs 0.612	0.129 vs 0.095	1,778 / 21,300	3,701
$\geq$ 1000 UMIs summed over the condition	0.919 vs 0.887	0.244 vs 0.103	1,778 / 21,300	660
$\geq$ 10 UMIs in both constructions	0.941 vs 0.925	0.406 vs 0.244	1,742 / 20,866	490
$\geq$ 100 UMIs in both constructions	0.968 vs 0.951	0.414 vs 0.164	1,191 / 12,337	34
$\geq$ 1000 UMIs in both constructions	0.997 vs 0.991	0.600 vs 0.112	271 / 2,924	0

2 of another (Table 5, Fig. 1b). Across the 114 mutants the median per-gene r is 0.612, so the own-culture ceiling is 0.729 by test-retest and 0.653 by decomposition (Fig. 1c). The two differ because the 114 paired mutants are enriched for strong responders, while the decomposition takes its signal variance from all 1,480. These numbers replace the void 0.775. The stored record averages both cultures where two exist; hypothesis, untested, that its ceiling is higher than the single-culture one.

Table 5: Kemmeren 2014, culture 1 against culture 2. A culture's profile is the mean over its dye-swapped arrays of the within-array $\log_2$(deletion/reference pool), computed by the loader's own functions from the GEO arrays of GSE42527 and GSE42526. The null pairs culture 1 of one mutant with culture 2 of another. Per-gene ceilings average over genes with at least three paired mutants; the decomposition takes the noise variance from the paired cultures and the total variance from the first culture of every mutant.

Kemmeren 2014 own-culture statistic	Value
Mutants with two cultures, n (responsive / non-responsive series)	114 of 1,480 (72 / 42)
Per-strain r across genes, median [IQR]	0.730 [0.602, 0.835]
Per-strain r, different mutants (null), median	0.284
Per-gene r across mutants, median [IQR]	0.612 [0.431, 0.754] (6,169 genes)
Ceiling, test-retest: mean over genes of $\sqrt{r_g}$	0.729
Ceiling, decomposition: mean over genes of $\sqrt{1-\sigma^2_{e,g}/\sigma^2_g}$	0.653
Void 019 cross-study ceiling (Sameith pair, identical arrays)	0.775

Hu 2007. No replicate statistic is computable. The paper says "Each experiment was independently carried out at least twice" (paper.md line 55), and the replicates were combined before release: the group "performed weighted duplicate averaging with custom Java software" (line 59). GEO GSE4654 holds one X score per strain; the per-array data went to the Longhorn Array Database and a lab site that was never archived. The only available bound is the agreement with Kemmeren, a median per-genotype r of 0.107 over 255 deletions and a per-gene r of 0.097 (Table 7). That agreement is a floor on Hu's reproducible signal; no ceiling is measured.

Boocock 2025 and N'Guessan 2025. Neither panel has a replicate: every pooled record sums the cells of one segregant (a median of 17 for Boocock Ap, 3 for N'Guessan), and the per-cell counts are not stored, so no split-half is possible (Table 6). No segregant appears in two panels: on a 5 kb grid of called parents the best concordance of any of the 393 pooled segregants with any of 27,744 de novo BY × RM cells is a median of 0.672, and with any of the 2,951 N'Guessan segregants 0.657, where unrelated segregants share half the genome and identity would be near 1; none reaches 0.95 (Fig. 2b). A weaker test still asks whether genotype carries expression across panels. For each pooled segregant, the 20 de novo cells nearest in genotype (mean concordance 0.647) agree with its deviation from the panel mean at a median r of 0.113, against 0.070 for 20 random cells; the paired difference is 0.040 (388 segregants, 63% positive, Wilcoxon $p = 1 \times 10^{-8}$). The single nearest N'Guessan segregant gives 0.101 against 0.060 (381 segregants, $p = 0.009$). At the panel level the summed BY × RM profiles agree at $r = 0.872$ between Boocock's pooled and de novo panels and 0.693 between Boocock's pooled panel and N'Guessan, which grew in YPD rather than minimal medium. Genotype posteriors differ by panel: a median 0% of markers of a pooled Boocock segregant and 1.7% of an N'Guessan one are called with posterior below 0.9, against 24 to 35% for a de novo cell.

3.2 Between datasets ✗

Records are matched by perturbed gene names and environment (isolates by strain name), and each pair is scored per shared genotype, across shared genes, and per shared gene, across shared genotypes (Table 7, Fig. 3; `experiments/043-expression-proteome-morphology/scripts/cross_dataset_correlation.py`, slurm jobs 5890 and 5892). Every pair compares log against log, so the base of a logarithm does not change a correlation.

Table 6: The segregant panels. A pooled record sums the cells matched to one previously genotyped segregant; a de novo record is one cell genotyped from its own transcripts. The posterior column is the marker-weighted fraction of a genotype whose called parent has posterior below 0.9.

Panel	Records, n	Cells, n	Cells per record, median	UMIs per record, median	Genes detected per record, median	Marker fraction with posterior < 0.9, median
Boocock Ap, pooled BY × RM	393	7,124	17	27,902	3,595	0.000
Boocock A, de novo BY × RM	27,744	27,744	1	4,017	1,228	0.236
Boocock B, de novo YJM145 × YPS163	44,784	44,784	1	2,130	859	0.352
Boocock 3004, de novo CBS2888 × YJM981	6,595	6,595	1	1,982	605	0.297
N'Guessan, pooled BY × RM	2,951	14,263	3	18,221	2,592	0.017

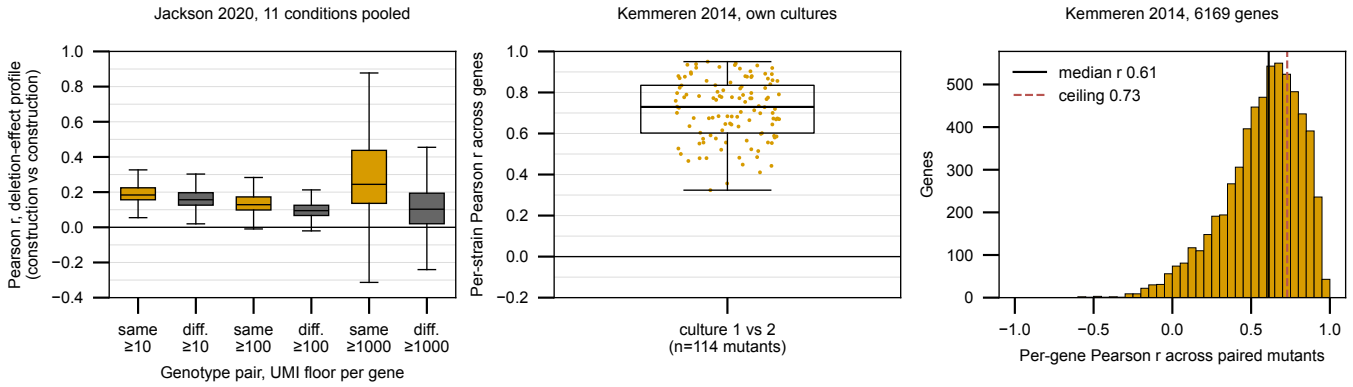


Figure 1: Two datasets measured against themselves. **a**, Jackson 2020: Pearson over genes of the deletion-effect profiles of two constructions in one condition, same genotype (orange) against different genotypes (gray), at three floors on a gene's UMIs summed over the condition; 1,778 and 21,300 pairs per floor, 11 conditions pooled, HO pairs left out. **b**, Kemmeren 2014: Pearson across genes between the two cultures of each of the 114 mutants profiled twice; culture 1 of one mutant against culture 2 of another gives a median of 0.284 (not drawn). **c**, per-gene Pearson across the 114 mutants for 6,169 genes, with the median r and the ceiling, the mean over genes of $\sqrt{\max(r_g, 0)}$.

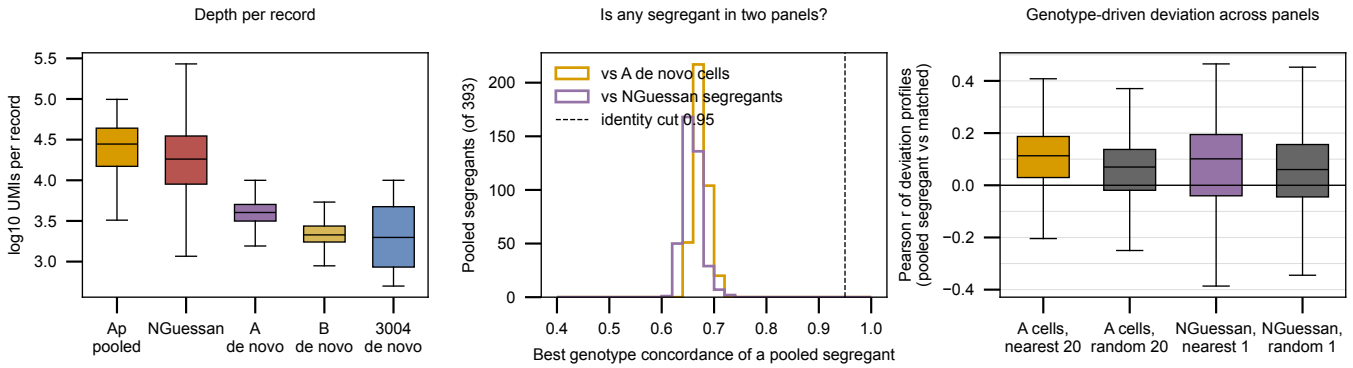


Figure 2: The segregant panels have no shared segregant and a weak shared genotype signal. **a**, UMIs per record by panel (pooled records sum a segregant's cells, de novo records are single cells). **b**, for each of the 393 pooled Boocock segregants, the best genotype concordance with any de novo cross-A cell (orange) or any N'Guessan segregant (purple), on a 5 kb grid of called parents; the dashed line marks 0.95. **c**, Pearson over genes between a pooled segregant's deviation from its panel and that of its 20 nearest de novo cells, 20 random cells, its nearest N'Guessan segregant, or a random one; genes with at least 10 UMIs on both sides.

Table 7: Agreement between datasets of the same readout type. Per genotype: Pearson and Spearman across the shared genes (at least 20). Per gene: Pearson across the shared genotypes (at least 5). Jackson is turned into a deletion effect against HO in the same environment; Messner and Zelezniak are log ratios to their record reference; Caudal and Muenzner are centered per gene over the isolates. The Jackson environment row pairs one genotype across two environments (605 genotype-environment pairs).

Pair	Shared genotypes, n	Per-genotype Pearson r, median [IQR]	Per-genotype Spearman, median	Per-gene Pearson r, median (genes, n)	Genes with per-gene r > 0.3, fraction
Kemmeren × Sameith Sm	82	1.000 [1.000, 1.000]	1.000	1.000 (6,169)	1.000
Kemmeren × Hu	255	0.107 [0.013, 0.184]	0.082	0.097 (5,803)	0.102
Kemmeren × Nadal, YPD	913	0.002 [−0.017, 0.022]	0.007	0.017 (5,891)	0.007
Kemmeren × Nadal, NaCl	920	−0.001 [−0.021, 0.022]	0.005	0.015 (5,896)	0.006
Kemmeren × Jackson, YPD 10 h	11	0.043 [0.026, 0.070]	0.047	0.035 (5,340)	0.247
Kemmeren × Jackson, YPD 4 h (a)	11	0.066 [0.027, 0.210]	0.030	0.118 (5,353)	0.321
Kemmeren × Jackson, YPD 4 h (b)	11	0.076 [0.036, 0.149]	0.045	0.052 (5,601)	0.253
Nadal YPD × Nadal NaCl	2,952	0.208 [0.168, 0.231]	0.043	0.183 (6,134)	0.050
Nadal duplicates, one condition	105	0.376 [0.323, 0.452]	0.178	0.062 (5,897)	0.089
Jackson, environment × environment	605	0.010 [−0.005, 0.029]	0.009	n/a	n/a
Messner × Zelezniak (protein)	89	0.049 [−0.016, 0.185]	0.050	0.085 (715)	0.067
Kemmeren mRNA × Messner protein	1,349	0.031 [−0.025, 0.112]	0.034	0.094 (1,832)	0.033
Caudal mRNA × Muenzner protein	759	0.208 [0.109, 0.307]	0.176	0.160 (1,572)	0.183
Muenzner turnover × abundance	55	−0.006 [−0.053, 0.047]	−0.003	−0.007 (1,120)	0.066

Sameith 2015 single mutants are Kemmeren 2014's arrays. All 82 Sameith Sm profiles are value-for-value identical to Kemmeren's for the same deletions (median per-genotype $r = 1.000$), and the GEO deposits show why. GSE42536 holds 144 Sameith single-mutant arrays: 115 are byte-identical re-deposits of GSE42527 or GSE42526 arrays, and the other 29 match a Kemmeren array exactly on the normalized Cy5 and Cy3 signal columns. Sameith's Methods defer to the earlier paper: "Full details of all gene expression profiling procedures have been described before [46]". The pair is therefore no replicate, and the experiment 019 expression ceiling built on it (test-retest 0.744 per strain, ceiling 0.77) is void; Table 5 replaces it. The 143 Sameith double-mutant arrays are new hybridizations.

Nadal-Ribelles 2025. Nadal-Ribelles agrees with Kemmeren at a median per-genotype r of 0.002 in YPD (913 deletions) and with itself not much better. Experiment 028 measured the same genotype across batches at a median r of 0.043 against 0.022 for different genotypes, and split halves at 0.080 against 0.051. The 0.376 of within-condition duplicates in Table 7 is Pearson driven by a few extreme log fold changes (the Spearman of the same pairs is 0.178); hypothesis, untested, that they come from genes with near-zero counts on one side.

Jackson 2020 across environments. One TF deletion's effect profile in one environment agrees with the same deletion's profile in another at a median r of 0.010 (605 pairs). The paper reports that TF deletions change far fewer genes than conditions do (96 differentially expressed genes in YPD, 160 in rapamycin), so a near-zero cross-environment agreement is what the paper leads one to expect; it does not show the data to be noise, and the within-condition construction pairs above measure the noise directly.

mRNA against protein. Deletions agree weakly: Kemmeren against Messner gives a median per-genotype r of 0.031 over 1,349 deletions and a per-gene r of 0.094 over 1,832 genes. Isolates agree more: Caudal against Muenzner gives 0.208 over 759 isolates and 0.160 per gene. Protein varies far more across isolates than across deletions (Fig. 4, left, x axes); hypothesis, untested, that this larger range is why the two readouts agree more there. Turnover against abundance is near zero (55 isolates).

Expression to morphology. A ridge from each gene-level matrix to the 281 CalMorph features of the shared Ohya genotypes (5-fold out-of-fold, one row-permutation null) is the one cross-label link that matters for a held-out probe (Table 8, Fig. 4, right). Kemmeren carries the most signal per genotype; Messner reaches nearly as far on three times the genotypes.

Table 8: Ridge from a gene-level matrix to CalMorph. Out-of-fold Pearson per CalMorph feature across held-out genotypes; inputs and targets z-scored and clipped at 8, penalty chosen on an inner split. The 116 moving features are the experiment 019 list (`calmorph_moving_features.json` on the multimodal-phenotype-retrospective worktree).

Input matrix	Genotypes shared with Ohya, n	Mean r over 281 features	Row-permutation null, mean r	Median r over 116 moving features
Kemmeren 2014	1,437	0.238	−0.005	0.348
Messner 2023	4,279	0.179	0.008	0.278
Hu 2007	259	0.114	−0.052	0.235
Sameith 2015 Sm	81	−0.014	−0.077	−0.085

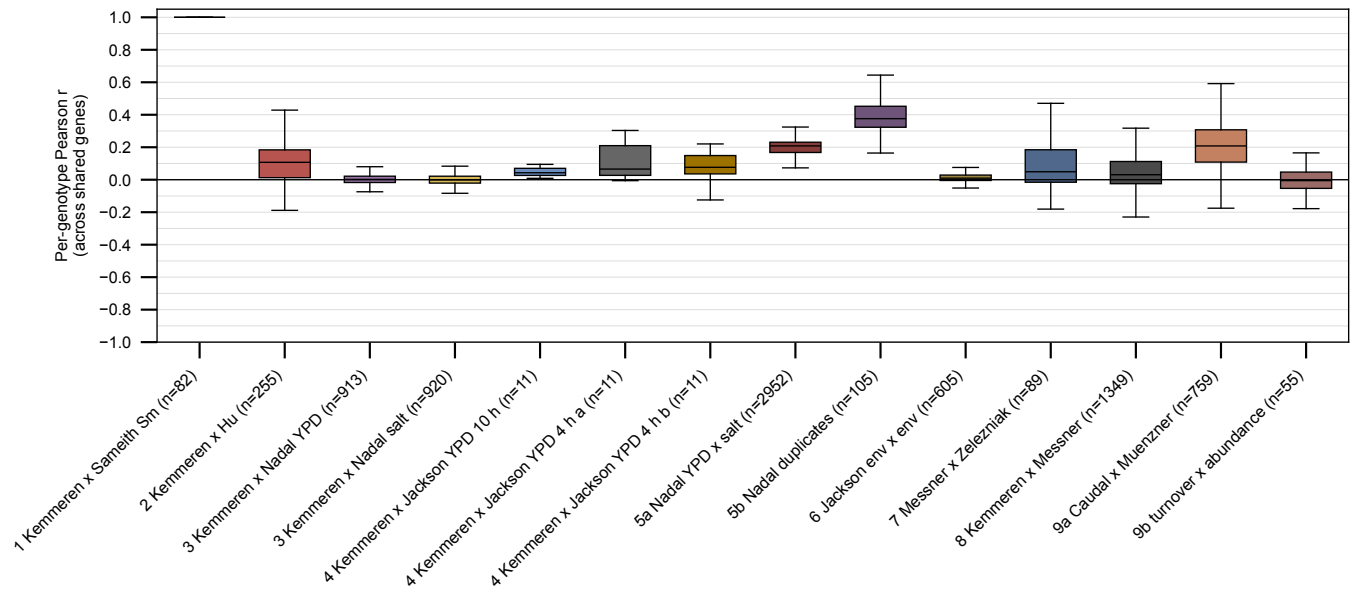


Figure 3: Agreement between datasets is low except for the Sameith copy. Per shared genotype, Pearson across the shared genes for every pair of Table 7; boxes are quartiles, whiskers 1.5 IQR, outliers not drawn.

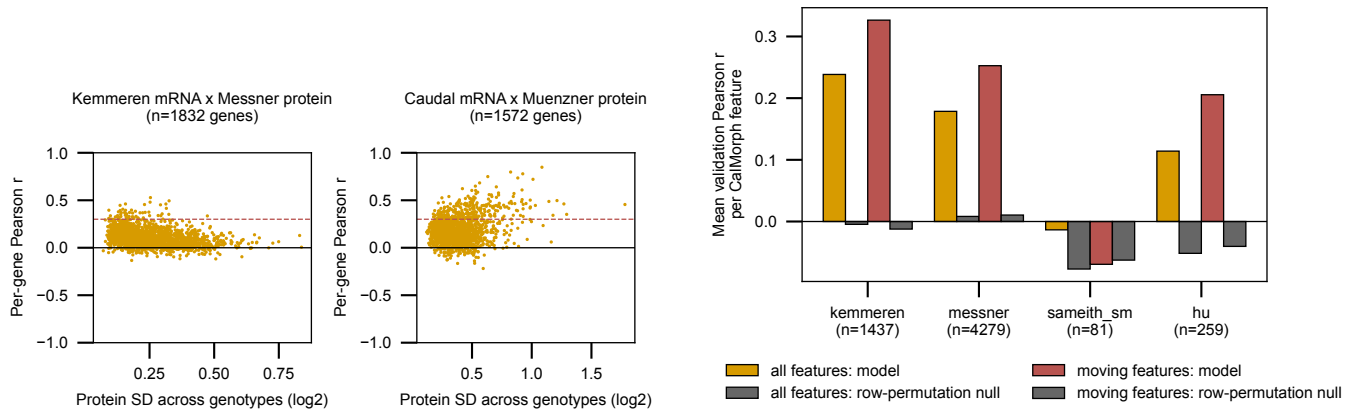


Figure 4: mRNA against protein, and expression to morphology. Left two panels: per-gene Pearson between mRNA and protein across the shared genotypes, against the protein's SD across those genotypes, for 1,832 genes over the deletions of Kemmeren and Messner and 1,572 genes over the isolates of Caudal and Muenzner; the dashed line marks $r = 0.3$. Right: mean out-of-fold Pearson per CalMorph feature of a ridge from each input matrix (Table 8), with its row-permutation null.

4 Which genotypes carry which labels ✗

A multitask model learns a shared representation only where one genotype carries more than one label. The census keys a genotype by its perturbed gene names, an isolate by its strain and a segregant by cross and identifier, and counts the genotypes each label and each pair of labels reach (Tables 9 and 10).

Table 9: Genotypes per label. Distinct genotype keys over every dataset carrying the label; Vaishnav is not keyed by gene and is left out.

Label	Genotypes, n
umi_count	82,479
protein_abundance	5,279
calmorph	4,678
expression_log2_ratio	3,685
expression_tpm	943
expression_score	262
degradation_rate	55

Table 10: Co-location. Genotypes carrying both labels, in any datasets. A label name can span datasets: `expression_log2_ratio` is Kemmeren, Sameith and Nadal-Ribelles; `umi_count` is Jackson, Boocock and N'Guessan; `protein_abundance` is Messner, Zelezniak and Muenzner.

Labels carried by the same genotype	Genotypes, n
expression_log2_ratio × calmorph	3,473
protein_abundance × calmorph	4,287
expression_log2_ratio × protein_abundance	3,317
expression_tpm × protein_abundance	759
umi_count × calmorph	12
expression_score × calmorph	259
isolates: Caudal × Muenzner protein	759
isolates: all three isolate datasets	55

Deletions. Morphology is the label most often shared. Of the 4,678 CalMorph genotypes, 4,287 also carry a protein profile and 3,473 an expression log ratio; the expression-by-protein overlap is 3,317 genotypes. That count includes Nadal-Ribelles, whose within-dataset agreement is near its null (Sec. 3.1); Kemmeren alone shares 1,349 genotypes with Messner. Kemmeren reaches Ohya on 1,437 genotypes and Messner on 4,279, the populations of the morphology ridge (Table 8).

Isolates. The isolate datasets are nested: all 55 turnover isolates are in the protein set and in Caudal's transcriptomes, and 759 of the 795 protein isolates have a Caudal transcriptome. No isolate carries a CalMorph profile.

Segregants and the rest. No segregant shares a key with any other dataset, and Boocock and N'Guessan share no segregant with each other (0 shared identifiers; Sec. 3.1 tests genotype identity rather than names). The 12 `umi_count` genotypes that carry morphology are Jackson's 11 transcription-factor deletions and the HO control.

5 Proposal: training batches ✗

What follows is a proposal for the group to accept, change or reject; nothing in it is decided. Each choice cites the measurement behind it (Table 11).

Table 11: Proposed batches. Records count the raw store; for Boocock only the pooled panel is proposed.

Batch	Datasets	Records, n	Probe target
1, deletions	Kemmeren 2014, Sameith 2015 Dm, Messner 2023	6,178	Ohya 2005 CalMorph, held out (4,678 records)
2, isolates and segregants	Caudal 2024, Muenzner 2024 protein, Boocock 2025 pooled (393 records), N'Guessan 2025	5,082	none named

Batch 1, deletions. Kemmeren 2014, Sameith 2015 Dm and Messner 2023 share one genotype representation, a set of deleted genes that the `Perturbation` processor already encodes, and each has a measured ceiling: 0.729 for Kemmeren's own cultures, 0.696 by decomposition over 388 wild-type replicates for Messner. Sameith Sm is left out because its 82 profiles are Kemmeren's; the Sameith doubles add 72 double deletions profiled by the same procedures (Sameith Methods defer to Kemmeren). Ohya 2005 CalMorph is the held-out probe target: it carries no input to the model, 4,287 of its genotypes have a Messner profile and 1,437 a Kemmeren one, a ridge on those inputs already reaches a mean r of 0.238 (Kemmeren) and 0.179 (Messner) over 281 features against nulls near zero, and its own ceiling is 0.611 over 122 wild-type replicates. A probe on the learned representation can then be read against both the ridge and that ceiling.

Batch 2, isolates and segregants. Caudal 2024 and Muenzner 2024 protein share 759 isolates and agree more than the deletions do (median per-isolate r 0.208 against 0.031). Boocock's 393 pooled segregants and N'Guessan's 2,951 are pseudobulk records of genome-sequenced BY × RM segregants, 7,124 and 14,263 cells, with near-certain genotypes (posterior below 0.9 on a median 0% and 1.7% of markers). The de novo Boocock cells are left out: single cells with medians of 2,000 to 4,000 UMIs, with 24 to 35% of its markers called below 0.9.

Both classes need a gene-level genotype the processor does not yet build. The proposed definite assumption: per gene, a segregant is marked at the genes whose block was inherited from the non-reference parent (RM, or the non-S288C parent of a cross), and blocks with posterior below a threshold such as 0.9 are dropped rather than guessed; an isolate is marked by a rule over perturbation types (loss of function, absence and presence, with sequence variants either dropped or restricted), since marking every variant gene sets 5,575 of 6,607 nodes. The probabilistic representation, a per-gene probability of the non-reference allele carried as a soft mask, is the later step. Whether the allele-marking assumption carries signal is open: the measured genotype signal across segregant panels is small (deviation r 0.113 against 0.070 for the nearest cells, Sec. 3.1).

Held back. Each of the following has a measured reason.

- Nadal-Ribelles 2025: the same genotype agrees across batches at a median r of 0.043 against 0.022 for different genotypes (028), and with Kemmeren at 0.002.
- Jackson 2020: two constructions of one genotype agree on the deletion effect at 0.129 against 0.095 for different genotypes (floor 100); 12 genotypes in 11 conditions add environments, not genotypes.
- Hu 2007: no replicate is released, and its only bound, agreement with Kemmeren, is 0.107 per genotype; the X score is not a log ratio.
- Zelezniak 2018: 97 kinase deletions, agreement with Messner 0.049 per genotype over 715 proteins, though its decomposition ceiling is 0.916.
- Vaishnav 2022: one scalar per promoter sequence and no gene-level genotype.
- Muenzner 2024 turnover: 55 isolates, and its agreement with abundance is near zero (-0.006 per isolate).